Alexander Kai Chen

+1 (858) 735-0720 | alexander.kai.chen@gmail.com | Dallas, TX, United States
http://alxv07.github.io | https://linkedin.com/in/alexanderkaichen | https://github.com/AlxV07

Mission

Exploring the intersection of computer science and classics to understand how language encodes thought. Building explainable systems to make reasoning transparent.

Publications

2025 — iOS as Acceleration
NeurIPS 2025 International Conference Efficient Reasoning Workshop

- Built novel distributed computing framework for using handheld phones as local accelerators
- Paper (https://arxiv.org/abs/2512.22180) published in the NeurIPS 2025 ER Workshop
- Democratizing AI; harnessing ubiquitous, underutilized devices in low-resource environments

C.S. & Classics Projects

2026 — august-mt: NeSy MT with Small LLMs for Low-resource Languages
Research Project, In-progress paper

- WIP: developing a neurosymoblic machine translation system combining LLMs and symbolic verificiation
- Translating Latin texts using LLM textual flexibility with formal grammatical rule construction; addressing training corpus limitations

2026 — Arwen: Verifiable Philosophical Argument Analysis
Research Project

- In-progress project: generating comparable "argument trees" using formal logic representations and language models
- Human-verifiably identify argument constructions, dependencies, fallacies in philosophical texts, debate transcripts

Research & Engineering Skills

- **Programming Languages:** Python, Java, Swift, C++, Lisp (Clojure), Prolog, Lean4
- **AI/ML:** Neural-Symbolic Systems, Explainable AI, Distributed Computing & Parallelism, Edge & Mobile Computing
- **Theory:** Discrete Mathematics, Formal Verification, Theorem Proving, Data Structures & Algorithms
- **Frameworks & Tools:** AWS, Azure, Angular, FastAPI, PyTorch, Mathlib
- **Workflows:** SCRUM, Agile, Agentic Orchestration

Experience

2025-2026 — SWE ET Internship
State Farm Enterprise Technology Division

- Summer 2025, 2026: Software Engineer Intern @ State Farm, Enterprise Technology division; AWS, Azure, Angular; SCRUM + Agile workflows
- Experience working with large and small internal teams on products deployed in high volume real-world settings

2024 — Explainable AI Education Research Internship
UTD Applied Logic & Programming Systems Lab (ALPS)

- Summer 2024: Explainable AI educational research intern @ UT Dallas Applied Logic & Programming Systems lab, under Professor Gopal Gupta
- Explored modeling scenarios using Prolog, combining neural networks and symbolic logic systems (NeSy), implemented logic problem solvers

Honors & Achievements

2025-2026 — ISEF State Fair
Texas Science & Engineering Fair

- Placed 2nd in "Software Design / Systems Software" category, earning full scholarship to Governor's Science & Technology Champion's Academy summer research program

2025-2026 — ISEF Regional Fair
Dallas Regional Science & Engineering Fair

- Placed 1st, winning "Software Design / Systems Software" category with project "MobilePipe: iPhones as Parallel Compute Accelerators for Local Machine Learning", advanced to State Fair.

2024-2025 — USACO Gold
USA Computing Olympiad, Competitive Algorithmic Programming

- Placed 1,022/12,591 world-wide in first contest in USA Computing Olympiad; advanced to Silver division in first year, Gold division following year

2022-2025 — Speech & Debate
NCFCA Speech & Debate, Coolidge Foundation Debate

- Communication, reasoning, and public presentation skills
- Placed 2nd team policy speaker nation-wide at NCFCA online national open tournament
- Ranked as 2nd individual speaker at Coolidge Foundation national debate tournament

2022-2024 — FRC Robotics World Championships
FIRST Robotics Competition

- Served on robotics programming team & led scouting data analytics application development, advancing to world final championship for two years

Education

2023–Present — Highschool
Homeschool

- 4.0 GPA, 1600 SAT, 6x APs score 5, currently completing Junior year

2025–Present — Dual Credit College Courses
Collin College

- Discrete Math (Grade: A, 100%)
- Linear Algebra (in-progress)

2023–Present — Online College Courses
Online

- "Data Structures & Algorithms Specialization" @ UCSD
- "Build a Modern Computer from First Principles" @ HUJI

2023–Present — C.S. & A.I.
Self-Directed study, mentored

- Focus areas include Agentic LRMs, NLP, Distributed Learning, XAI
- Working with college professors on research projects

2023–Present — Classical Languages & Philosophy
Highschool Classical Education, Self-Directed study

- Studying classical, ecclesiastical Latin, Greek and Roman philosophy.